Paper OOL-1: The Thermodynamic Mandate and Dissipative Structuring: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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Abstract

This opening manuscript frames abiogenesis as a nonequilibrium systems problem rather than a single lucky molecular accident. Constraint-Driven Flux Dynamics (CDFD) is used as a coarse-grained language for asking when environmental energy flux, geometric constraint, active surface formation, and memory-like persistence can reinforce one another. The central object is the adaptive ratio

$$\Psi_s = \left(\frac{\Phi}{C}\right) SM_s,$$

where Φ denotes available driving flux, C denotes the effective resistance or constraint field, S denotes responsive organization, and M_s denotes retained structural memory. The paper does not prove that life is inevitable in every driven environment. It establishes the dependency structure that later papers test in geochemical, polymerization, compartment, replication, parasitic, photochemical, and synthesis settings.

1 Introduction

Living systems maintain local order by coupling themselves to external gradients. The origin-of-life problem is therefore not only the problem of making particular molecules; it is also the problem of sustaining useful work against dilution, hydrolysis, side reactions, and spatial loss. In CDFD language, a prebiotic system becomes interesting when a flux source and a constraint architecture create a persistent opportunity for organization.

This manuscript supplies the series-level vocabulary. This vocabulary is CDFL, the CDFD flow-constraint-memory language used by the public runtime and by the manuscript series. It deliberately keeps the claim at the level of mechanism class: driven systems can favor structures that increase usable dissipation, but only if those structures also retain enough organization to persist.

2 Scientific Lineage and Complementarity

Part II stands on a long origin-of-life tradition rather than replacing it. Oparin's coacervate program, Wächtershäuser's surface-metabolism proposal, Russell and Martin's alkaline hydrothermal-vent work, Eigen and Orgel's error-threshold logic, Kauffman's autocatalytic-set theory, Szostak's protocell program, Lane's bioenergetic arguments, Powner's nucleotide chemistry, Patel's cyanosulfidic network chemistry, and Damer's cycling-environment hypothesis all identify real parts of the puzzle [12, 10, 5, 2, 7, 3, 11, 4, 9, 8, 1]. CDFD's role is to provide a single dependency grammar across those traditions: what is the input, what constrains it, what routes it, what is retained, and what would falsify the claimed coupling?

This framing is meant to complement those schools, not erase their disagreements. Vent models, hot-spring models, mineral-surface chemistry, RNA-world chemistry, metabolism-first chemistry, and protocell experiments can all be compared by asking whether they solve the same sequence of functional gates.

Status: Historical framing

3 Relationship to Part I

Part I of this series introduced CDFL, the CDFD state grammar for flow, constraint, responsiveness, memory, and threshold crossing in fundamental-physics language [6]. Part II does not require the reader to accept the stronger physical claims of Part I. It uses the shared notation as a disciplined bookkeeping layer: Φ names drive, C names resistance or constraint, S names responsive organization, and M_s names retained structure. The biological question is whether those variables can be instantiated by geochemical, chemical, and protocellular mechanisms.

Status: Framing

4 Flux, Constraint, Surface, and Memory

The shared phenomenological balance used throughout this Part II sequence is

$$\frac{\partial \Phi}{\partial t} = \nabla \cdot \left(\frac{1}{C} \nabla \Phi \right) + S - D, \tag{1}$$

where D collects loss, degradation, washout, and uncoupled dissipation. The terms are interpreted as follows:

- **Flux (Φ)**: chemical, redox, photochemical, thermal, or electrochemical drive available to the system.
- **Constraint (C)**: transport resistance, activation barrier, geometric confinement, permeability, or maintenance burden.
- **Responsiveness (S)**: the ability of local structure to route flux into productive pathways rather than noise.

- **Systemic memory (M_s):** persistence of structure, sequence, composition, topology, or boundary state across time.

All symbols are local phenomenological variables unless explicitly tied to a measurement. They should not be read as universal constants. Later papers redefine specialized symbols only after stating their physical meaning, because the same letter can otherwise mean different things in reaction kinetics, transport theory, and bioenergetics.

The adaptive ratio

$$\Psi_s = \left(\frac{\Phi}{C}\right) S M_s \tag{2}$$

is not introduced as a universal constant. It is a bookkeeping device for tracking whether the components needed for persistence are co-present. A high flux without retention burns out. A strong constraint without flux freezes. A responsive surface without memory remains transient.

5 Necessary Conditions for the Part II Model

The Part II claim holds only if four necessary conditions are met together. They are stated explicitly because origin-of-life narratives often over-weight one ingredient: a favored molecule, a favored setting, or a favored energy source. CDFD treats those ingredients as incomplete unless they satisfy a coupled gate.

1. **Sustained disequilibrium.** The environment must supply repeatable drive, not a single pulse. Operationally, Φ must remain positive over the relevant renewal interval.

2. **Finite constraint.** The system must be neither freely mixed nor sealed off. In symbols, $0 < C < \infty$: too little constraint gives washout, while too much constraint prevents exchange and renewal.

3. **Responsive routing.** The local structure must alter product distributions or retention. A surface, pore, droplet, or polymer has CDFD significance only when it changes where flux goes, represented by $S > 0$.

4. **Retained memory.** Some product, pattern, boundary state, or catalytic bias must survive later destructive phases. The working test is $M_s(t + nT) > M_s(t)$ over at least one repeated environmental sequence.

Status: Derived release guardrail

6 Dependency-Gate Diagnostic

The final release adds a lightweight script for this opening paper, `supplementary_ool_01.py`. The script scans a two-dimensional grid of drive Φ and constraint C, then computes Eq. (2) using bounded phenomenological functions for S and M_s. The result is not a simulation of life. It is a necessary-condition map showing where the opening notation is subcritical, near-critical, or overloaded.

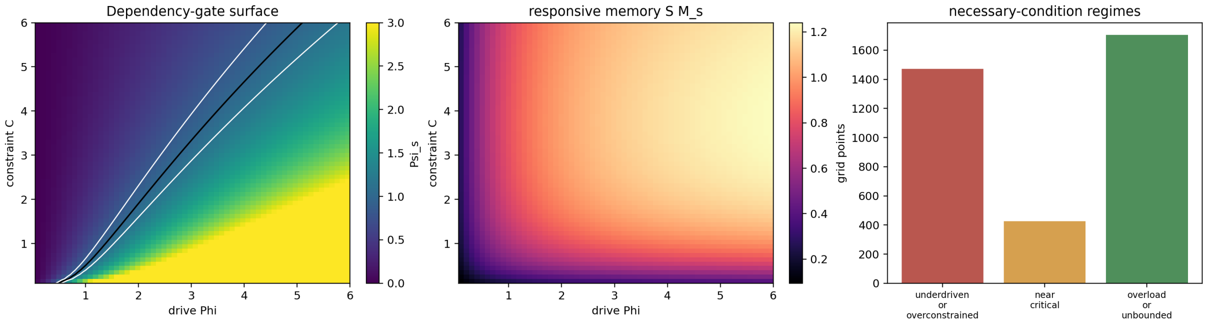

Figure 1: Dependency-gate diagnostic for Paper 1. The near-critical band marks where drive, finite constraint, responsive routing, and retained memory are co-present in the toy surface. It is an interpretive map for the twelve-paper chain, not evidence that abiogenesis has occurred.

The scan reports 3,600 grid points, of which 425 fall in the near-critical window $0.85 \leq \Psi_s \leq 1.15$. That number has no universal meaning. Its purpose is to force the release to distinguish three regimes:

- underdriven or overconstrained systems, where chemistry cannot build persistent memory;
- near-critical systems, where later mechanisms may matter;
- overloaded systems, where excess drive outruns retention or control.

Status: Derived diagnostic

7 Dissipative Structuring

Driven systems can form structures that route energy more effectively than homogeneous diffusion. In prebiotic settings, the relevant question is whether such structures can become chemically consequential: mineral interfaces, pores, films, droplets, polymers, and reaction loops all change the effective relationship between Φ and C.

For a bare chemical soup, one may have $S \approx 0$ and $M_s \to 0$, so transient fluctuations do not accumulate. A successful origin pathway must raise both quantities without disconnecting from the external drive. This is the handoff to Paper 2: the framework needs geological boundary conditions before it can become a concrete origin-of-life argument.

8 Claim Boundary

The strongest defensible claim of this paper is conditional:

> If a prebiotic environment supplies sustained disequilibrium, spatial constraint, and chemistry capable of retaining useful structure, then CDFD predicts selection for configurations that improve coupled throughput and persistence.

This is not a proof that a particular vent, pool, mineral, or molecule was historically necessary. The later papers narrow those possibilities by adding Fe–S redox chemistry, distributed mineral transport, proton/ion coupling, cycling polymerization, autocatalysis, chemical alphabets, compartments, replication thresholds, parasitic failure modes, and photochemical expansion.

9 Series-Level Falsification Logic

The opening paper can fail in several ways. If prebiotic experiments show that product persistence is independent of constraint architecture, then C is a weak or decorative variable. If surfaces change flux but not endpoint chemistry, then S is not a useful organizing parameter. If repeated cycles do not improve retained product distributions relative to matched steady controls, then M_s does not describe chemical memory in the relevant setting. Finally, if each paper in the series requires a different incompatible definition of Ψ_s, the CDFL grammar fails as a unifying language even if individual origin-of-life mechanisms remain interesting.

Status: Falsifiable framing

10 Conclusion

The thermodynamic opening move is a dependency graph, not a victory lap. Life-like organization requires flux, constraint, responsive routing, and retained memory to reinforce one another. Paper 2 therefore begins the concrete chain with the most immediate physical substrate: iron–sulfur redox and mineral constraint scaffolds.

11 Reproducibility Note

The companion script `supplementary_ool_01.py` writes `dependency_gate_surface.png` under `outputs/paper_01/`, plus a grid CSV, a necessary-conditions CSV, and summary JSON. The output is a necessary-condition visualization, not validation of abiogenesis.

References

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